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Unit – 5 : Session – 4 : SLO – 1 | Symbolic Programming Concepts

■ I. Questions & Answers

1. Symbolic computation — computation in which mathematical objects (variables, expressions, equations) are manipulated as exact symbols rather than approximate numbers, e.g. solving $x^2 - 1 = 0$ to get exactly $x = 1$ and $x = -1$.
2. Python library — SymPy is the standard Python library for symbolic computation (alternatives: SageMath; Octave/Mathematica outside Python).
3. Example of a symbolic equation — $\text{Eq}(x^2 + 2x + 1, 0)$, i.e. $x^2 + 2x + 1 = 0$; SymPy solves it exactly: $\text{solve}(x^2 + 2x + 1, x) \rightarrow [-1]$.